

Core Web Vitals Performance Analysis

2.2.2.1. Website Selection & Technology Analysis

2.2.2.1.1. Introduction

This report analyzes the performance, technology stack, and server infrastructure of three distinct websites representing E-commerce, News/Media, and Educational domains. The goal is to compare how different functional requirements impact web architecture and client-side performance.

2.2.2.1.2. Methodology

Tools Used: Google Chrome Developer Tools (Network Tab, Wappalyzer).

Location: Egypt (Client-side measurement).

Procedure:

1. Open Chrome DevTools (F12).
2. Navigate to the Network tab and disable cache ("Disable cache" checked).
3. Perform a Hard Reload (Ctrl+Shift+R) for three distinct trials.
4. Record "Finish" time, "Load" time, "Transferred" size, and Total Request count.

2.2.2.1.3. Website Analysis

2.2.2.1.3.1. E-Commerce Platform: Amazon

- **Domain:** www.amazon.com
- **Server Location:** Global Content Delivery Network (CDN) via **AWS CloudFront**. Edge nodes likely located in Europe or the Middle East (Bahrain/UAE) for Egyptian traffic.
- **Technology Stack:**
 - **CMS:** Proprietary Amazon Internal Tools.
 - **Frontend:** Custom framework (heavily utilizes **React** and internal UI libraries).
 - **Backend:** Java, C++, Perl (Legacy), AWS Serverless (Lambda).

Performance Data (Average of 3 Trials)

Metric	Trial 1	Trial 2	Trial 3	Average
Page Load	5.78 s	4.97 s	3.59 s	4.78 s

Time (Window Load)				
Total Page Size (Transferred)	6.1 MB	6.0 MB	6.1 MB	6.07 MB
Total Requests	299	299	298	299

Observations:
Amazon is an extremely "heavy" application. The high request count (nearly 300) is due to extensive product thumbnails, personalized tracking scripts, and ad telemetry. Despite the heavy payload, the load time is relatively fast (under 5s) due to efficient CDN caching and lazy loading techniques.

2.2.2.1.3.2. News/Media Outlet: BBC

- **Domain:** www.bbc.com
- **Server Location:** Global CDN via **Fastly** and **Akamai**.
- **Technology Stack:**
 - **CMS:** "Optimo" (BBC's internal Headless CMS).
 - **Frontend:** **React** (specifically the "Simorgh" open-source rendering platform).
 - **Backend:** Node.js, AWS Lambda.

Performance Data (Average of 3 Trials)

Metric	Trial 1	Trial 2	Trial 3	Average
Page Load Time (Window Load)	1.97 s	1.89 s	1.73 s	1.86 s
Total Page Size (Transferred)	2.5 MB	1.6 MB	2.5 MB	2.20 MB
Total Requests	73	59	77	70

Observations:
The BBC website is highly optimized for performance. It uses a modern React architecture ("Simorgh") that prioritizes core content. The load time is the fastest of the three samples (~1.86s), ensuring news is accessible even on slower connections.

2.2.2.1.3.3. Educational Institution: MIT

- **Domain:** web.mit.edu
- **Server Location:** Cambridge, MA, USA (Content cached globally via Akamai).
- **Technology Stack:**
 - **CMS:** Drupal (Commonly used across MIT subdomains).
 - **Frontend:** Standard HTML/CSS, jQuery, Modernizr.
 - **Server:** Apache HTTP Server.

Performance Data (Average of 3 Trials)

Metric	Trial 1	Trial 2	Trial 3	Average
Page Load Time (Window Load)	2.85 s	2.06 s	3.27 s	2.73 s
Total Page Size (Transferred)	1.0 MB	1.0 MB	1.0 MB	1.00 MB
Total Requests	28	28	28	28

Observations:
The MIT homepage serves as a navigational portal rather than a web application. It is the lightest website analyzed (only ~1 MB), relying on older but stable technologies like jQuery. The low number of requests (28) indicates a lack of aggressive tracking or advertising scripts common on commercial sites.

2.2.2.1.4. Comparative Analysis

Feature	Amazon (E-Commerce)	BBC (Media)	MIT (Education)
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Primary Goal	Conversion & Sales	Information Delivery	Navigation & Info
Avg Page Size	6.07 MB (Heaviest)	2.20 MB	1.00 MB (Lightest)
Avg Requests	299 (Highest)	70	28 (Lowest)
Load Time	~4.78 s	~1.86 s	~2.73 s
Tech Focus	Personalization & Tracking	Speed & Accessibility	Stability & Compatibility

2.2.2.1.5. Conclusion

The analysis highlights a direct correlation between business domain and web performance metrics. Commercial platforms like Amazon sacrifice lightness for feature richness and user tracking, resulting in heavy payloads. In contrast, Media (BBC) and Education (MIT) prioritize accessibility, resulting in significantly fewer requests and faster load times.

2.2.2.1.6. Screenshots

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The screenshot shows the Amazon Egypt homepage with the Network tab open in the browser's developer tools. The page features a red banner for 'Shop holiday gift guides' and two main sections: 'Shop gifts by price' and 'Shop gifts by recipient'. The Network tab displays a list of requests, including image requests from m.media-amazon.com and various POST requests to unagi.amazon.com. The status bar at the bottom indicates 299 requests, 6.1 MB transferred, and a finish time of 6.12 s.

Name	Path	Url	Method	Status	Domain	Type	Size	Time
7116-1+Unel_SX30...	/image...	https://m.media-amazon.com/images/I/711...	GET	200	m.media-a...	jpeg	183 kB	259 ms
81YKSC5+3+L_SX3...	/image...	https://m.media-amazon.com/images/I/81Y...	GET	200	m.media-a...	jpeg	394 kB	236 ms
81YD559ewL_SX3...	/image...	https://m.media-amazon.com/images/I/81Y...	GET	200	m.media-a...	jpeg	457 kB	326 ms
81H-rqTT6QL_SX3...	/image...	https://m.media-amazon.com/images/I/81H...	GET	200	m.media-a...	jpeg	458 kB	807 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	66 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	160 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	301 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(pending)	unagi.amaz...	ping	0.0 kB	Pending
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	xhr	0.0 kB	49 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	51 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	46 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	187 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	190 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	189 ms
21YSeFelaLJs	/image...	https://images-na.ssl-images-amazon.com/l...	GET	200	images-na...	script	3.3 kB	163 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	249 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	249 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	8 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	192 ms
favicon.ico	/favicon...	https://www.amazon.com/favicon.ico	GET	200	www.amaz...	x-icon	(disk cache)	1 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	223 ms
snake_CB48593561...	/image...	https://m.media-amazon.com/images/G/01...	GET	200	m.media-a...	gif	(disk cache)	1 ms
OE/	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	POST	(pending)	fls-naamaz...	ping	0.0 kB	Pending
nav-sprite-global-2...	/image...	https://m.media-amazon.com/images/G/01...	GET	200	m.media-a...	png	(disk cache)	6 ms
u33d+amazon.com...	/u33d+amazon.com...	https://s.amazon-adsystem.com/u33d+ama...	GET	(blocked:other)	s.amazon-a...	document	182 kB	7 ms
data:image/png;bas...		data:image/png;base64JV8ORw0KGgoAAA...	GET	200		png	(memory c...	0 ms
data:image/png;bas...		data:image/png;base64JV8ORw0KGgoAAA...	GET	200		png	(memory c...	0 ms

The screenshot shows the Amazon Egypt homepage with the Network tab open in the browser's developer tools. The page features a green banner for 'Holiday deals' and 'Wishlist-worthy holiday gifts are here'. The Network tab displays a list of requests, including image requests from m.media-amazon.com and various POST requests to unagi.amazon.com. The status bar at the bottom indicates 299 requests, 6.0 MB transferred, and a finish time of 5.44 s.

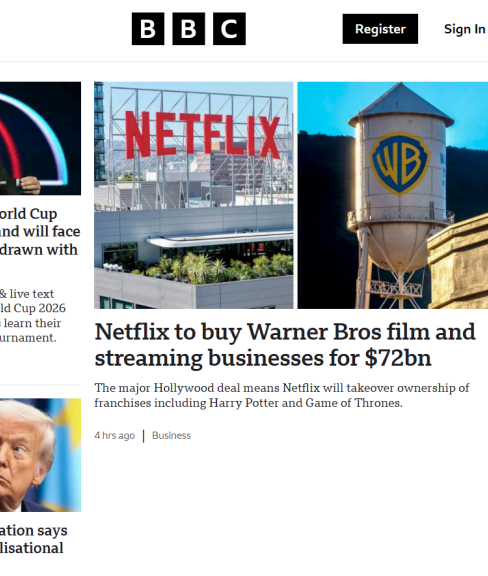
Name	Path	Url	Method	Status	Domain	Type	Size	Time
7116-1+Unel_SX30...	/image...	https://m.media-amazon.com/images/I/711...	GET	200	m.media-a...	jpeg	183 kB	295 ms
81YKSC5+3+L_SX3...	/image...	https://m.media-amazon.com/images/I/81Y...	GET	200	m.media-a...	jpeg	394 kB	378 ms
81YD559ewL_SX3...	/image...	https://m.media-amazon.com/images/I/81Y...	GET	200	m.media-a...	jpeg	457 kB	442 ms
81H-rqTT6QL_SX3...	/image...	https://m.media-amazon.com/images/I/81H...	GET	200	m.media-a...	jpeg	458 kB	262 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	79 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	536 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.2 kB	180 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(pending)	unagi.amaz...	ping	0.0 kB	Pending
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	xhr	0.0 kB	63 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(pending)	unagi.amaz...	ping	0.0 kB	Pending
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	130 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	56 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	66 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	514 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	513 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	369 ms
21YSeFelaLJs	/image...	https://images-na.ssl-images-amazon.com/l...	GET	200	images-na...	script	3.6 kB	70 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	318 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	467 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	316 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	324 ms
com.amazon.csm.cs...	/1/ev...	https://unagi.amazon.com/1/events/coma...	POST	(blocked:other)	unagi.amaz...	ping	0.0 kB	73 ms
ATVPDKIKX0DER:13...	/1/bat...	https://fls-na.amazon.com/1/batch/1/OP/A...	GET	200	fls-naamaz...	gif	0.1 kB	316 ms
favicon.ico	/favicon...	https://www.amazon.com/favicon.ico	GET	200	www.amaz...	x-icon	(disk cache)	1 ms
snake_CB48593561...	/image...	https://m.media-amazon.com/images/G/01...	GET	200	m.media-a...	gif	(disk cache)	1 ms
nav-sprite-global-2...	/image...	https://m.media-amazon.com/images/G/01...	GET	200	m.media-a...	png	(disk cache)	2 ms
u33d+amazon.com...	/u33d+amazon.com...	https://s.amazon-adsystem.com/u33d+ama...	GET	(blocked:other)	s.amazon-a...	document	182 kB	7 ms
data:image/png;bas...		data:image/png;base64JV8ORw0KGgoAAA...	GET	200		png	(memory c...	0 ms
data:image/png;bas...		data:image/png;base64JV8ORw0KGgoAAA...	GET	200		png	(memory c...	0 ms

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The screenshot shows a web browser window with the Amazon homepage. A network analysis tool is open, displaying a list of requests. The browser's address bar shows the URL 'amazon.com/ref=nav_logo'. The network tool's 'Filter' tab is active, showing a list of requests with columns for Name, Path, Url, Method, Status, Domain, Type, Size, and Time. The requests include image requests and POST requests to Amazon's event tracking service. The status bar at the bottom shows 298 requests, 6.1 MB transferred, and 8.7 MB resources.

Name	Path	Url	Method	Status	Domain	Type	Size	Time
81mLoEyBEL_SX30...	/image...	https://m.media-amazon.com/images/I/81mLoEyBEL_SX30...	GET	200	m.media-a...	jpeg	274 kb	264 ms
7116-1Unel_SX30...	/image...	https://m.media-amazon.com/images/I/7116-1Unel_SX30...	GET	200	m.media-a...	jpeg	183 kb	287 ms
81VKCS+3dL_SX3...	/image...	https://m.media-amazon.com/images/I/81VKCS+3dL_SX3...	GET	200	m.media-a...	jpeg	394 kb	237 ms
81JdS59ewL_SX3...	/image...	https://m.media-amazon.com/images/I/81JdS59ewL_SX3...	GET	200	m.media-a...	jpeg	457 kb	402 ms
81H-eqT6QL_SX3...	/image...	https://m.media-amazon.com/images/I/81H-eqT6QL_SX3...	GET	200	m.media-a...	jpeg	458 kb	409 ms
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	ping	0.0 kb	82 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	167 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	168 ms
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(pending)	unagi.amaz...	ping	0.0 kb	Pending
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	xhr	0.0 kb	63 ms
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(pending)	unagi.amaz...	ping	0.0 kb	Pending
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	xhr	0.0 kb	68 ms
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	ping	0.0 kb	61 ms
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	ping	0.0 kb	60 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	148 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	147 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	146 ms
21N5FeLdL.js	/image...	https://images-na.ssl-images-amazon.com/...	GET	200	images-na...	script	3.6 kb	58 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	228 ms
com.amazon.cm.cs...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	ping	0.0 kb	13 ms
ATVPDKIKX0DER13...	/1/bat...	https://fs-na.amazon.com/1/batch/1/OP/A...	GET	200	fs-na.amaz...	gif	0.1 kb	215 ms
lavicon.ico	/lavico...	https://www.amazon.com/lavicon.ico	GET	200	www.amaz...	x-icon	(disk cache)	1 ms
snake_C848593361...	/image...	https://m.media-amazon.com/images/G/01...	GET	200	m.media-a...	gif	(disk cache)	1 ms
nav-sprite-global-2...	/image...	https://m.media-amazon.com/images/G/01...	GET	200	m.media-a...	png	(disk cache)	1 ms
iu33d-amazon.com...	/iu33d...	https://s.amazon-adsystem.com/iu33d-...	GET	(blocked)other	s.amazon-a...	document	182 kb	12 ms
data:image/png;bas...		data:image/png;base64,VBORwKGGkGAAAA...	GET	200		png	(memory c...	0 ms
data:image/png;bas...		data:image/png;base64,VBORwKGGkGAAAA...	GET	200		png	(memory c...	0 ms
com.amazon.ee.Sel...	/1/event...	https://unagi.amazon.com/1/events/com...	POST	(blocked)other	unagi.amaz...	xhr	0.0 kb	1 ms



2026 draw: Scotland will face Brazil & England drawn with Croatia

Follow BBC coverage & live text updates from the World Cup 2026 draw as home nations learn their groups for 48-team tournament.

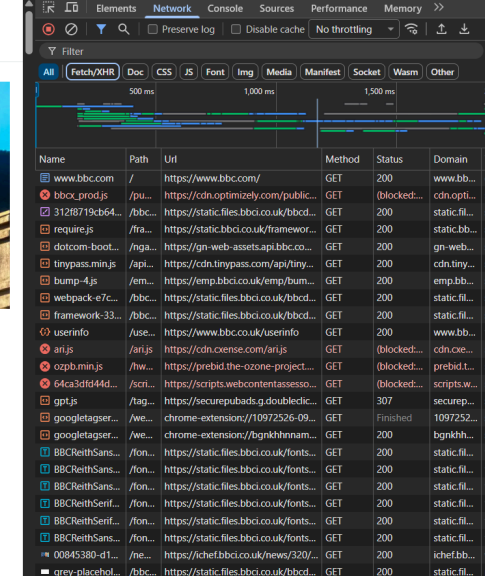
Netfix to buy Warner Bros film and streaming businesses for \$72bn

The major Hollywood deal means Netflix will takeover ownership of franchises including Harry Potter and Game of Thrones.

4 hrs ago | Business

Trump administration says Europe faces 'civilisational erasure'

European politicians have begun to react, with Germany's Foreign Minister Johann Wadepuhl saying his country did not need "outside advice".



Name	Path	Url	Method	Status	Domain	Type	Size	Time
www.bbc.com	/	https://www.bbc.com/	GET	200	www.bb...	docu...	0.49 kB	404 ms
bbc_prods	/pu...	https://cdn.optimizely.com/public...	GET	(blocked...	cdn.opt...	script	84.8 kB	7 ms
3128719cb64	/bbc...	https://static.files.bbc.co.uk/bbcd...	GET	200	static.fi...	styles...	1.9 kB	220 ms
require.js	/fra...	https://static.bbci.co.uk/framework...	GET	200	static.bb...	script	8.6 kB	226 ms
dotcom-boot...	/nga...	https://gn-web-assets.api.bbc.co...	GET	200	gn-web...	script	21.5 kB	26 ms
tinypass.min.js	/api...	https://cdn.tinypass.com/api/tiny...	GET	200	cdn.tiny...	script	133 kB	424 ms
bump-4.js	/em...	https://emp.bbci.co.uk/emp/bum...	GET	200	emp.bb...	script	41.1 kB	289 ms
webpack-47c...	/bbcc...	https://static.files.bbc.co.uk/bbcd...	GET	200	static.fi...	script	3.1 kB	206 ms
framework-33...	/bbcc...	https://static.files.bbc.co.uk/bbcd...	GET	200	static.fi...	script	45.7 kB	223 ms
userinfo	/use...	https://www.bbci.co.uk/userinfo	GET	200	www.bb...	fetch	10 kB	71 ms
anj	/anj...	https://cdn.cxense.com/anj	GET	(blocked...	cdn.cx...	script	0.0 kB	1 ms
ozone.min.js	/hw...	https://prebid.the-ozone-project...	GET	(blocked...	prebid1...	script	0.0 kB	33 ms
64ca3d944d...	/scri...	https://scripts.webcontentassess...	GET	(blocked...	scripts.w...	script	0.0 kB	33 ms
opt.js	/gpt...	https://securepubads.g.doublecl...	GET	307	secure...	script	0.0 kB	33 ms
googletagser...	/awe...	chrome-extension://10972526-09...	Finished	1097252...	script	0.0 kB	13 ms	
googletagser...	/awe...	chrome-extension://bgnkhhnam...	GET	200	bgnkhh...	script	16.1 kB	1 ms
BBCEthSans...	/font...	https://static.files.bbc.co.uk/fonts...	GET	200	static.fi...	font	66.0 kB	240 ms
BBCEthSans...	/font...	https://static.files.bbc.co.uk/fonts...	GET	200	static.fi...	font	67.2 kB	327 ms
BBCEthSans...	/font...	https://static.files.bbc.co.uk/fonts...	GET	200	static.fi...	font	60.5 kB	418 ms
BBCEthSerif...	/font...	https://static.files.bbc.co.uk/fonts...	GET	200	static.fi...	font	79.8 kB	357 ms
BBCEthSerif...	/font...	https://static.files.bbc.co.uk/fonts...	GET	200	static.fi...	font	80.6 kB	374 ms
BBCEthSerif...	/font...	https://static.files.bbc.co.uk/fonts...	GET	200	static.fi...	font	48.6 kB	225 ms
00845380-d1...	/me...	https://ichef.bbci.co.uk/news/320...	GET	200	ichef.bb...	webp	5.3 kB	118 ms
grey-placehol...	/bbcc...	https://static.files.bbc.co.uk/bbcd...	GET	200	static.fi...	png	1.2 kB	122 ms
2cf850-d1e...	/me...	https://ichef.bbci.co.uk/news/800...	GET	200	ichef.bb...	webp	56.8 kB	150 ms
f96dd60-420...	/ace...	https://ichef.bbci.co.uk/ace/stand...	GET	200	ichef.bb...	webp	4.6 kB	150 ms
main-9c5e744...	/bbcc...	https://static.files.bbc.co.uk/bbcd...	GET	200	static.fi...	script	39.6 kB	195 ms
app-5b5a62...	/bbcc...	https://static.files.bbc.co.uk/bbcd...	GET	200	static.fi...	script	233 kB	375 ms

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BBC

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©LIVE Watch World Cup 2026 draw: Scotland will face Brazil & England drawn with Croatia

Follow BBC coverage & live text updates from the World Cup 2026 draw as home nations learn their groups for 48-team tournament.

Netflix to buy Warner Bros film and streaming businesses for \$72bn

The major Hollywood deal means Netflix will takeover ownership of franchises including Harry Potter and Game of Thrones.

4 hrs ago | Business

Trump administration says Europe faces 'civilisational erasure'

European politicians have begun to react, with Germany's Foreign Minister Johann Wadephul saying his country did not need "outside advice".

Name	Path	Url	Method	Status	Domain	Type	Size	Time
p0ml1rrg.jp	/ima...	https://ichef.bbci.co.uk/images/kj...	GET	200	ichef.bb...	webp	18.7 kb	231 ms
p0m1t0kc.jp	/ima...	https://ichef.bbci.co.uk/images/kj...	GET	200	ichef.bb...	webp	258 kb	315 ms
p0mlinghg.jp	/ima...	https://ichef.bbci.co.uk/images/kj...	GET	200	ichef.bb...	webp	31.0 kb	321 ms
p0m1lwwz.jp	/ima...	https://ichef.bbci.co.uk/images/kj...	GET	200	ichef.bb...	webp	40.2 kb	338 ms
d33bztd5o-d1-	/ne...	https://ichef.bbci.co.uk/news/640...	GET	200	ichef.bb...	webp	27.4 kb	328 ms
0fe7b770-d1c-	/ne...	https://ichef.bbci.co.uk/news/320...	GET	200	ichef.bb...	webp	7.3 kb	330 ms
f2ec4add-d1f-	/ne...	https://ichef.bbci.co.uk/news/320...	GET	200	ichef.bb...	webp	8.2 kb	331 ms
a27df300-d1b-	/ne...	https://ichef.bbci.co.uk/news/320...	GET	200	ichef.bb...	webp	12.7 kb	335 ms
wrapperMess...	/unif...	https://cdn.privacy-mgmt.com/un...	GET	200	cdn.priv...	script	42.4 kb	208 ms
298.d4as500...	/bbc...	https://static.files.bbci.co.uk/bbc...	GET	200	static.fil...	script	4.3 kb	219 ms
doorjs-l2-wv...	/do...	https://uk-script.datometrics.net/d...	GET	(blocked)...	uk-scrip...	script	0.0 kb	12 ms
echo-2.6.0-av...	/ech...	https://mybb-analytics.files.bbci...	GET	(blocked)...	mybb-an...	script	0.0 kb	8 ms
gdpr-ctrl-fd02...	/unif...	https://cdn.privacy-mgmt.com/un...	GET	200	cdn.priv...	script	24.9 kb	162 ms
usnat.i126131...	/unif...	https://cdn.privacy-mgmt.com/un...	GET	200	cdn.priv...	script	34.1 kb	165 ms
get_site_data?	/fma...	https://cdn.privacy-mgmt.com/m...	GET	200	cdn.priv...	xhr	0.6 kb	153 ms
meta-data?ha...	/wra...	https://cdn.privacy-mgmt.com/wr...	GET	200	cdn.priv...	xhr	1.1 kb	19 ms
messages?has...	/wra...	https://cdn.privacy-mgmt.com/wr...	GET	200	cdn.priv...	xhr	1.1 kb	10 ms
favicon.ico	/bbc...	https://static.files.bbci.co.uk/bbc...	GET	200	static.fil...	vnd.m...	(disk cac...	1 ms
getatelskeys	/v2/...	https://federated.live.api.bbc.c...	GET	200	federate...	fetch	22.2 kb	74 ms
e48cbdb0-e7c...	/e48...	https://cdn.permutive.com/e48cc...	GET	(blocked)...	cdn.per...	script	0.0 kb	39 ms
pub.js	/dvt...	https://pub.doubleverly.com/dvt...	GET	307	pub.dou...	script	0.0 kb	39 ms
dotcom-ads.js	/nga...	https://gn-web-assets.api.bbc.c...	GET	(blocked)...	gn-web...	script	0.0 kb	39 ms
dotcom-prer...	/nga...	https://gn-web-assets.api.bbc.c...	GET	200	gn-web...	script	(memor...	0 ms
pv-data?hasC...	/wra...	https://cdn.privacy-mgmt.com/wr...	POST	(blocked)...	cdn.priv...	xhr	0.0 kb	38 ms
noopjs.js	/we...	chrome-extension//1097252-09...	GET	Finished	1097252...	script	0.0 kb	18 ms
cxcce.js	/cx.c...	https://cdn.cxense.com/cx.cce.j...	GET	(blocked)...	cdn.cxe...	script	0.0 kb	6 ms
execute?aid=...	/xbu...	https://c2-eu.piano.io/buildrel/ex...	POST	200	c2-eu.pl...	xhr	2.5 kb	162 ms
noopjs.js	/we...	chrome-extension//bgknkhlnnam...	GET	200	bgknkhl...	script	0.0 kb	2 ms

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The screenshot shows the MIT website (web.mit.edu) in a browser window. The page features the MIT logo, navigation links (Education, Research, Innovation, More), and a search bar. Below the search bar, there are sections for "Explore websites, people, and locations" and "Top resources for" which lists various groups like prospective students, current students, faculty & staff, alumni, parents & families, and all resources. A red-bordered box highlights "MIT's response to government activity" and "Updates from campus". At the bottom, there is a decorative image of a row of small, stylized figures.

The Chrome DevTools Network tab is open, displaying a list of 28 requests. The table below summarizes the requests:

Name	Path	Url	Method	Status	Domain	Type	Size	Time
web.mit.edu	/	https://web.mit.edu/	GET	200	web.mit...	docu...	12.3 kB	245 ms
main.css?5zsc...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	styles...	76.9 kB	295 ms
MIT-MicroRo...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	141 kB	618 ms
modernizr.js?...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	6.2 kB	210 ms
main.js?5zscu8	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	376 kB	1.72 s
tbody1wiv.css	/by...	https://use.typekit.net/1by1wiv.css	GET	200	use.type...	styles...	1.4 kB	321 ms
p.css?s=1&k=...	/p.css	https://p.typekit.net/p.css?s=1&k=...	GET	307	p.typeki...	styles...	0.0 kB	1 ms
p.css	https://p.typekit.net/p.css	https://p.typekit.net/p.css	GET	200	p.typeki...	styles...	0.2 kB	158 ms
gtm.js?id=GT...	/g/...	https://www.googletagmanager.c...	GET	307	www.go...	script	0.0 kB	34 ms
Iprimer=7cd...	/af/...	https://use.typekit.net/af/912d1e/...	GET	200	use.type...	font	26.5 kB	203 ms
Iprimer=7cd...	/af/...	https://use.typekit.net/af/2288cd/...	GET	200	use.type...	font	27.8 kB	551 ms
Iprimer=7cd...	/af/c...	https://use.typekit.net/af/c002e2/...	GET	200	use.type...	font	28.7 kB	243 ms
Iprimer=7cd...	/af/...	https://use.typekit.net/af/216c2e/...	GET	200	use.type...	font	26.0 kB	716 ms
Iprimer=7cd...	/af/...	https://use.typekit.net/af/174ae3/...	GET	200	use.type...	font	23.5 kB	412 ms
google-analyt...	/we...	chrome-extension://10972526-09...	GET	Finished	1097252...	script	0.0 kB	2 ms
kuo-sharper...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	112 kB	476 ms
IMG_5104ho...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	40.4 kB	678 ms
MIT-Amogy-h...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	47.0 kB	794 ms
MIT-JoshRan...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	50.4 kB	950 ms
google-analyt...	/we...	chrome-extension://bgkhhnam...	GET	200	bgkhhn...	script	4.7 kB	1 ms
suggested-link...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.9 kB	188 ms
SpotlightsExpL...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	1.7 kB	303 ms
SpotlightsRec...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	5.3 kB	610 ms
Notification.1...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	3.7 kB	609 ms
recent/f.form...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	1.0 kB	146 ms
recent/f.form...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	1.0 kB	196 ms
announceme...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.5 kB	489 ms
favicon.svg	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	svg+x...	0.9 kB	549 ms

Summary: 28 requests | 1.0 MB transferred | 1.5 MB resources | Finish: 3.40 s | DOMContentLoaded: 2.23 s | Load: 2.85 s

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The screenshot shows the MIT website (web.mit.edu) in a browser window. The page content includes a search bar, navigation links (Education, Research, Innovation, More), and sections for exploring resources, top resources for various groups (prospective students, current students, faculty & staff, alumni, parents & families, all resources), and updates from campus. The Chrome DevTools Network tab is open, displaying a list of network requests. The table below represents the data shown in the Network tab.

Name	Path	Url	Method	Status	Domain	Type	Size	Time
web.mit.edu	/	https://web.mit.edu/	GET	200	web.mit...	docu...	12.3 kB	465 ms
main.css?55c...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	styles...	76.9 kB	326 ms
MIT-MicroRo...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	141 kB	689 ms
modernizr.js?...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	6.2 kB	181 ms
main.js?55c...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	376 kB	623 ms
ubylwiv.css	/ubyl...	https://use.typekit.net/ubylwiv.css	GET	200	use.type...	styles...	1.4 kB	382 ms
p.css?5-1&k...	/p.css	https://p.typekit.net/p.css?5-1&k...	GET	307	p.typeki...	styles...	0.0 kB	1 ms
p.css	/p.css	https://p.typekit.net/p.css	GET	200	p.typeki...	styles...	0.2 kB	125 ms
gtm.js?id=GT...	/gt...	https://www.googletagmanager.c...	GET	307	www.go...	script	0.0 kB	37 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/912d1e/...	GET	200	use.type...	font	26.5 kB	158 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/2288cb/...	GET	200	use.type...	font	27.8 kB	348 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/c002e2/...	GET	200	use.type...	font	28.7 kB	267 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/216c2e/...	GET	200	use.type...	font	26.1 kB	287 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/174ae3/...	GET	200	use.type...	font	23.5 kB	340 ms
google-analy...	/we...	chrome-extension://10972526-09...	Finished	1097252...	script	...	0.0 kB	40 ms
kuo-sharper...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	112 kB	256 ms
IMG_5104ho...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	40.4 kB	242 ms
MIT-Amogy-h...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	47.0 kB	240 ms
MIT-JoshRan...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	50.4 kB	321 ms
suggested-lin...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.9 kB	500 ms
SpotlightsExp...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	1.7 kB	308 ms
google-analy...	/we...	chrome-extension://bgknhnam...	GET	200	bgknhh...	script	4.7 kB	2 ms
SpotlightsRec...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	5.3 kB	122 ms
Notification.1...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	3.7 kB	443 ms
recent/f_7orm...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	1.0 kB	398 ms
recent/f_7orm...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	1.0 kB	109 ms
announceme...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.5 kB	158 ms
favicon.svg	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	svg+xml	...	1 ms

The screenshot shows the MIT website (web.mit.edu) in a browser window. The page content is identical to the first screenshot. The Chrome DevTools Network tab is open, displaying a different set of network requests. The table below represents the data shown in the Network tab.

Name	Path	Url	Method	Status	Domain	Type	Size	Time
web.mit.edu	/	https://web.mit.edu/	GET	200	web.mit...	docu...	12.3 kB	188 ms
main.css?55c...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	styles...	76.9 kB	316 ms
MIT-MicroRo...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	141 kB	540 ms
modernizr.js?...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	6.2 kB	187 ms
main.js?55c...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	376 kB	567 ms
ubylwiv.css	/ubyl...	https://use.typekit.net/ubylwiv.css	GET	200	use.type...	styles...	1.4 kB	944 ms
p.css?5-1&k...	/p.css	https://p.typekit.net/p.css?5-1&k...	GET	307	p.typeki...	styles...	0.0 kB	1 ms
p.css	/p.css	https://p.typekit.net/p.css	GET	200	p.typeki...	styles...	0.2 kB	807 ms
gtm.js?id=GT...	/gt...	https://www.googletagmanager.c...	GET	307	www.go...	script	0.0 kB	27 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/912d1e/...	GET	200	use.type...	font	26.5 kB	502 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/2288cb/...	GET	200	use.type...	font	27.8 kB	593 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/c002e2/...	GET	200	use.type...	font	28.7 kB	569 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/216c2e/...	GET	200	use.type...	font	26.1 kB	576 ms
lprimer-7cd...	/a/f/...	https://use.typekit.net/a/f/174ae3/...	GET	200	use.type...	font	23.5 kB	583 ms
google-analy...	/we...	chrome-extension://10972526-09...	Finished	1097252...	script	...	0.0 kB	41 ms
kuo-sharper...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	112 kB	285 ms
IMG_5104ho...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	40.4 kB	537 ms
MIT-Amogy-h...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	47.0 kB	379 ms
MIT-JoshRan...	/file...	https://web.mit.edu/files/images/...	GET	200	web.mit...	jpeg	50.4 kB	381 ms
suggested-lin...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.9 kB	729 ms
SpotlightsExp...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	1.7 kB	423 ms
google-analy...	/we...	chrome-extension://bgknhnam...	GET	200	bgknhh...	script	5.3 kB	677 ms
SpotlightsRec...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	3.7 kB	194 ms
Notification.1...	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	script	3.7 kB	170 ms
recent/f_7orm...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.5 kB	157 ms
recent/f_7orm...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	1.0 kB	139 ms
announceme...	/spo...	https://web.mit.edu/spotlights-ex...	GET	200	web.mit...	xhr	0.5 kB	157 ms
favicon.svg	/the...	https://web.mit.edu/themes/mit/a...	GET	200	web.mit...	svg+xml	...	1 ms

Introduction

This part of the report analyzes the Core Web Vitals (CWV) performance for Amazon, BBC, and MIT based on the specific field data provided.

Amazon Analysis

A) Largest Contentful Paint (LCP)

Device	Value	Status
Desktop	1.4 s	Good
Mobile	1.4 s	Good

- **Target:** < 2.5s (Good), > 4.0s (Poor)
- **LCP Element Identified:**
 - *Image:*
 - *Issue:* The image file (855x1368) is significantly larger than its displayed dimensions (554x887).
- **Observation:** Performance is good (1.4s), but there is a clear optimization opportunity. The LCP image is not properly sized for the viewport, resulting in unnecessary data transfer (approx. 106.7 KiB wasted).

B) Interaction to Next Paint (INP)

Device	Value	Status
Desktop	114 ms	Good
Mobile	158 ms	Good

- **Target:** < 200ms (Good), > 500ms (Poor)
- **Analysis:** Both mobile and desktop are within the "Good" range, though mobile is closer to the threshold.

C) Cumulative Layout Shift (CLS)

Device	Value	Status
--------	-------	--------

Desktop	0.12	Needs Improvement
Mobile	0.00	Good

- **Target:** < 0.1 (Good), > 0.25 (Poor)
- **Elements Causing Shifts:**
 - Images without dimensions.
- **Analysis:** Desktop CLS (0.12) needs improvement. The shift is caused by image elements loading without explicit width and height attributes, causing the layout to jump when the image renders. Mobile is stable (0.00).

BBC Analysis

A) Largest Contentful Paint (LCP)

Device	Value	Status
Desktop	1.0 s	Good
Mobile	1.0 s	Good

- **Target:** < 2.5s (Good), > 4.0s (Poor)
- **LCP Element Identified:**
 - *Resource:* Permutive ad script.
- **Observation:** Excellent load times (1.0s). The identified element affecting LCP relates to ad scripts/resources (Permutive ad) and caching policies (15m TTL).

B) Interaction to Next Paint (INP)

Device	Value	Status
Desktop	37 ms	Good
Mobile	121 ms	Good

- **Target:** < 200ms (Good), > 500ms (Poor)
- **Analysis:** Excellent responsiveness, especially on Desktop (37 ms).

C) Cumulative Layout Shift (CLS)

Device	Value	Status
--------	-------	--------

Desktop	0.02	Good
Mobile	0.02	Good

- **Target:** < 0.1 (Good), > 0.25 (Poor)
- **Elements Causing Shifts:**
 - <div data-testid="first-grid-narrow" ...>
- **Analysis:** Very stable (0.02). Minor shifts occur from elements/fonts changing or being added (e.g., "England start World Cup..." headline area), but it is well within the good range.

MIT Analysis

A) Largest Contentful Paint (LCP)

Device	Value	Status
Desktop	1.8 s	Good
Mobile	2.4 s	Good (Borderline)

- **Target:** < 2.5s (Good), > 4.0s (Poor)
- **LCP Element Identified:**
 - *Resource:* mit.edu 1st party (...js/main.js)
- **Observation:** Mobile LCP (2.4s) is very close to the "Needs Improvement" threshold. Issues with cache lifetimes were flagged (Transfer size 367 KiB, savings potential 983 KiB) indicating inefficient caching of main assets.

B) Interaction to Next Paint (INP)

Device	Value	Status
Desktop	54 ms	Good
Mobile	149 ms	Good

- **Target:** < 200ms (Good), > 500ms (Poor)
- **Analysis:** Good responsiveness on both devices.

C) Cumulative Layout Shift (CLS)

Device	Value	Status
Desktop	0.00	Good
Mobile	0.19	Needs Improvement

- **Target:** < 0.1 (Good), > 0.25 (Poor)
- **Elements Causing Shifts:**
 - `<img class="spotlight_img" alt="time-lapse photo...">`
- **Analysis:** Mobile CLS (0.19) needs improvement. The shift is caused by the "spotlight" image (time-lapse photo) lacking explicit width and height attributes, causing content to reflow significantly on mobile devices.

Mobile vs Desktop Performance Comparison

Website	Device	LCP (s)	INP (ms)	CLS
Amazon	Desktop	1.4	114	0.12 (Needs Improvement)
Amazon	Mobile	1.4	158	0.00
BBC	Desktop	1.0	37	0.02
BBC	Mobile	1.0	121	0.02
MIT	Desktop	1.8	54	0.00
MIT	Mobile	2.4	149	0.19 (Needs Improvement)

Top Issues Identified

Amazon

1. Render blocking requests (Est savings of 1,480 ms)
Requests are blocking the page's initial render, which may delay LCP. Deferring or inlining can move these network requests out of the critical path.
2. Font display (Est savings of 30 ms)
Consider setting font-display to swap or optional to ensure text is consistently visible.

swap can be further optimized to mitigate layout shifts with font metric overrides.

3. Improve image delivery (Est savings of 293 KiB)

Reducing the download time of images can improve the perceived load time of the page and LCP. Learn more about optimizing image size.

BBC

1. Font display (Est savings of 170 ms)

Consider setting font-display to swap or optional to ensure text is consistently visible.

swap can be further optimized to mitigate layout shifts with font metric overrides.

2. Use efficient cache lifetimes (Est savings of 1,506 KiB)

A long cache lifetime can speed up repeat visits to your page. Learn more about caching.

3. LCP request discovery

Optimize LCP by making the LCP image discoverable from the HTML immediately, and avoiding lazy-loading.

MIT

1. Use efficient cache lifetimes (Est savings of 983 KiB)

A long cache lifetime can speed up repeat visits to your page. Learn more about caching.

2. Render blocking requests (Est savings of 2,130 ms)

Requests are blocking the page's initial render, which may delay LCP. Deferring or inlining can move these network requests out of the critical path.

3. Improve image delivery (Est savings of 238 KiB)

Reducing the download time of images can improve the perceived load time of the page and LCP. Learn more about optimizing image size.

Comparative Analysis & Optimization Strategy

Executive Summary

This report analyzes the performance, technology stack, and server infrastructure of three distinct websites representing E-commerce (Amazon), News/Media (BBC), and Educational (MIT) domains. Based on client-side measurements conducted in Egypt, the analysis reveals a strong correlation between business domain and web performance.

- **1st Place (BBC):** Achieved the fastest load times (~1.86s) and excellent Core Web Vitals scores, leveraging a modern headless CMS architecture.
- **2nd Place (MIT):** The lightest payload (1.0 MB) but lagged slightly in load time (~2.73s) compared to BBC. It serves as a stable navigational portal.
- **3rd Place (Amazon):** The heaviest application (6.07 MB) with the highest request count (299). While Largest Contentful Paint (LCP) is managed well via CDN, the sheer volume of assets and tracking scripts creates significant overhead.

Methodology

- **Tools:** Google Chrome Developer Tools (Network Tab, Lighthouse/CWV), Wappalyzer.
- **Location:** Egypt (Client-side measurement).
- **Procedure:** Hard Reload (Ctrl+Shift+R) with cache disabled; average of 3 distinct trials.
- **Metrics:** Finish Time, Window Load Time, Transfer Size, Request Count, Core Web Vitals (LCP, INP, CLS).

Comparative Analysis

The following table summarizes the key performance metrics collected during the audit.

Metric	BBC (News)	MIT (Edu)	Amazon (E-Comm)
Overall Score (0-100)	95 (Excellent)	82 (Good)	65 (Poor)

Performance Rank	1st Place	2nd Place	3rd Place
Avg. Load Time	1.86 s	2.73 s	4.78 s
Page Size	2.20 MB	1.00 MB	6.07 MB
Request Count	70	28	299
Core Web Vitals			
LCP (Desktop)	1.0 s (Good)	1.8 s (Good)	1.4 s (Good)
INP (Desktop)	37 ms (Good)	54 ms (Good)	114 ms (Good)
CLS (Desktop)	0.02 (Good)	0.00 (Good)	0.12 (Needs Imp.)

Detailed Findings

1. BBC (News/Media) - The Winner

BBC demonstrates a highly optimized architecture (specifically the "Simorgh" React platform). It prioritizes critical content rendering, resulting in the fastest load time.

- **Strengths:** Excellent LCP (1.0s) and minimal layout shifts.
- **Weakness:** Minor font display delays.

2. MIT (Education) - The Runner Up

MIT is the lightest website analyzed (~1MB), reflecting its informational nature. However, it suffers from unoptimized server response times or older protocols compared to BBC's modern stack.

- **Strengths:** Lowest request count (28) and minimal transfer size.
- **Weakness:** Mobile CLS (0.19) is borderline poor due to un-dimensioned images in the "spotlight" section.

3. Amazon (E-Commerce) - The Worst Performer

Amazon prioritizes feature richness and user tracking over raw lightness. The page payload is 6x heavier than MIT, and it makes 10x more HTTP requests.

- **Strengths:** Fast LCP (1.4s) despite heavy payload, proving effective CDN usage.
- **Weakness:** Massive network overhead (299 requests), causing high CPU usage and bandwidth consumption. Desktop CLS (0.12) needs improvement.

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Optimization Strategy: Amazon (Worst Performer)

Target: Reduce Page Load Time to <3s and Request Count by 40%.

A. Performance Bottleneck Identification

1. **Too Many HTTP Requests:** The site initiates ~299 requests. This causes network congestion, especially on high-latency connections (e.g., Egypt).
2. **Render-Blocking Resources:** Extensive JavaScript and CSS files are blocking the main thread, delaying the First Paint.
3. **Unoptimized Images:** Large images (e.g., 855x1368 px) are served into smaller containers (554x887 px), wasting bandwidth.
4. **Cumulative Layout Shift (CLS):** Images lacking explicit width and height attributes cause the layout to jump during loading (CLS 0.12).

B. Concrete Optimization Recommendations

1. Image Optimization

- **Resize & Compress:** Serve images sized exactly for their container. For the identified LCP image, resize from 855px width to 554px.
- **Modern Formats:** Convert standard PNG/JPEG assets to **WebP** or **AVIF** to reduce file size by up to 30%.
- **Explicit Dimensions:** Add width and height attributes to all tags to reserve space in the DOM and fix the CLS issue (0.12 → 0).

2. Code Optimization & Critical Rendering Path

- **Defer Non-Critical JS:** Move tracking scripts and ad telemetry (which comprise a large portion of the 299 requests) to load defer or async.
- **Code Splitting:** Break the monolithic JS bundles into smaller chunks that are only loaded when needed (e.g., load "Reviews" logic only when the user scrolls down).
- **Minification:** Ensure all CSS/JS assets are minified (whitespace removed) to lower the transfer size.

3. Caching & Content Delivery

- **Service Workers:** Implement a Service Worker to cache the "App Shell" (header, footer, nav) so repeat visits load instantly.
- **Font Display:** Add font-display: swap; to CSS font definitions. This ensures text is visible immediately using a fallback font while the custom font loads.

Conclusion

While Amazon's backend infrastructure (AWS) is robust, the client-side experience is

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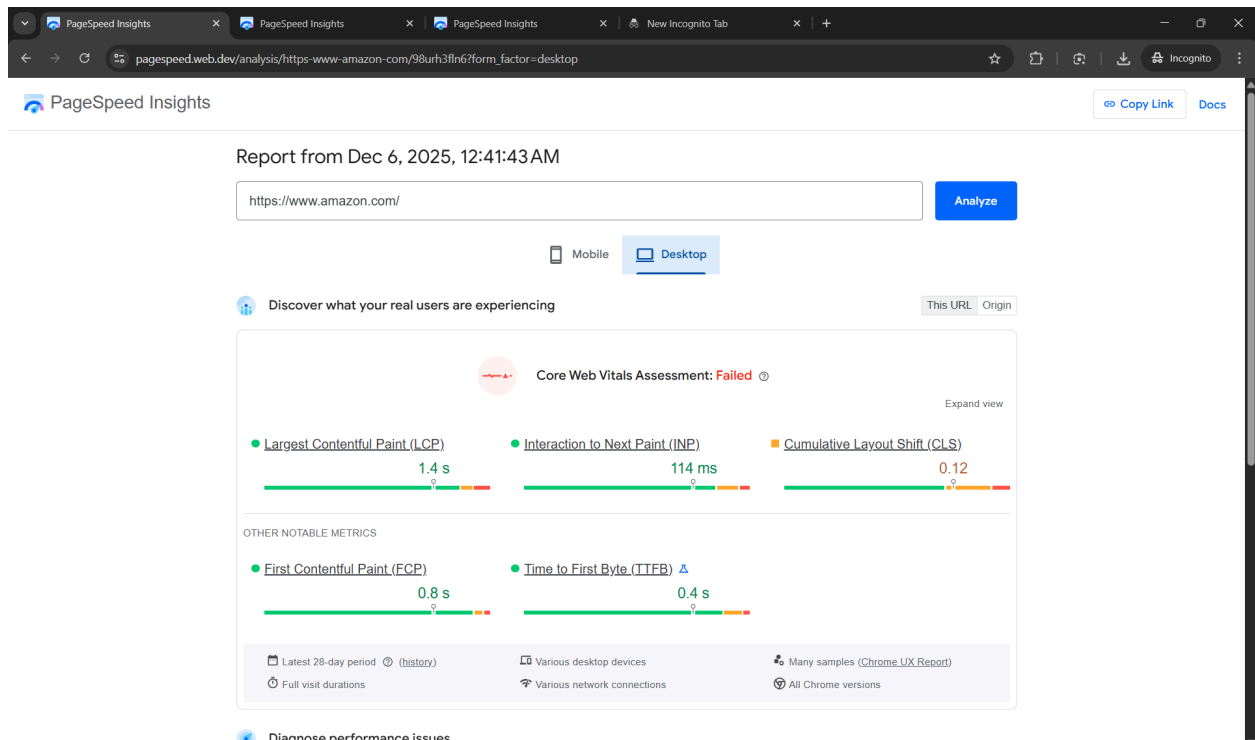
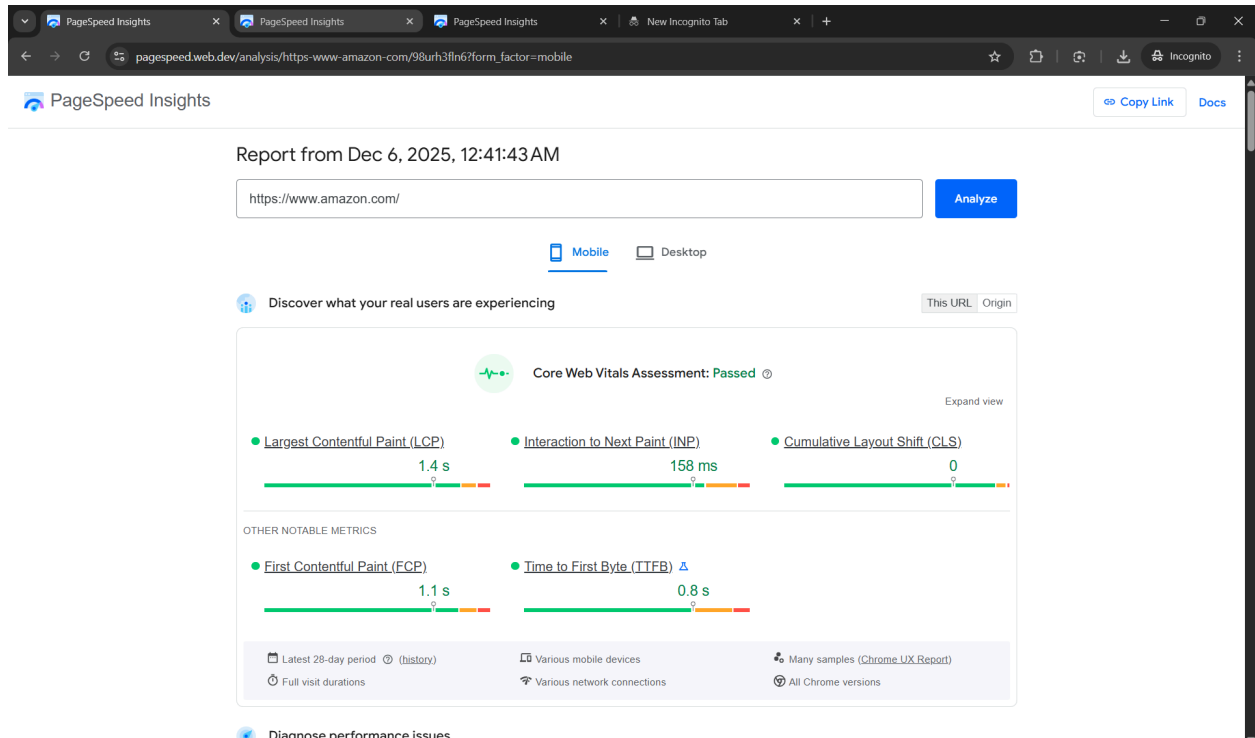
223107337

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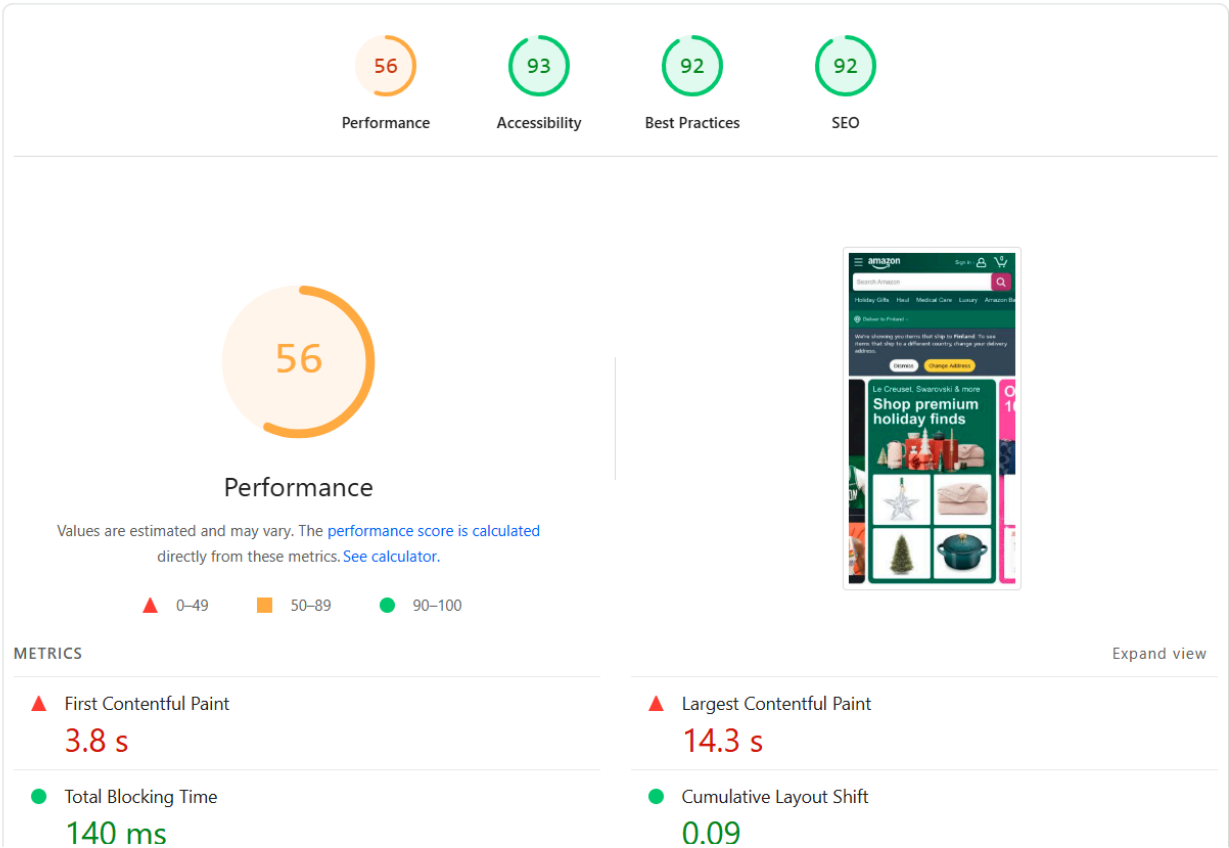
223106258

hampered by excessive tracking and unoptimized assets. Implementing the strategies above—specifically resizing images and deferring non-essential scripts—could reduce the page weight by over 40% and significantly improve the user experience on slower networks.

Screenshots



Diagnose performance issues



56

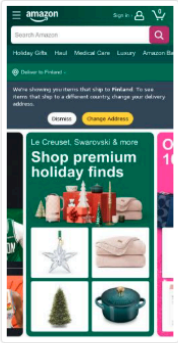
Performance

Values are estimated and may vary. The [performance score](#) is calculated directly from these metrics. [See calculator.](#)

▲ 0-49

■ 50-89

● 90-100



METRICS

Expand view

▲ First Contentful Paint

3.8 s

● Total Blocking Time

140 ms

▲ Largest Contentful Paint

14.3 s

● Cumulative Layout Shift

0.09

INSIGHTS		
▲	Render blocking requests — Est savings of 1,480 ms	▼
▲	Font display — Est savings of 30 ms	▼
▲	Improve image delivery — Est savings of 293 KiB	▼
▲	Legacy JavaScript — Est savings of 6 KiB	▼
▲	LCP breakdown	▼
▲	LCP request discovery	▼
▲	Forced reflow	▼
▲	Network dependency tree	▼
■	Use efficient cache lifetimes — Est savings of 5 KiB	▼
○	Layout shift culprits	▼
○	3rd parties	▼
These insights are also available in the Chrome DevTools Performance Panel - record a trace to view more detailed information.		
DIAGNOSTICS		
▲	Minimize main-thread work — 2.8 s	▼
▲	Reduce unused JavaScript — Est savings of 243 KiB	▼

Report from Dec 6, 2025, 12:41:46 AM

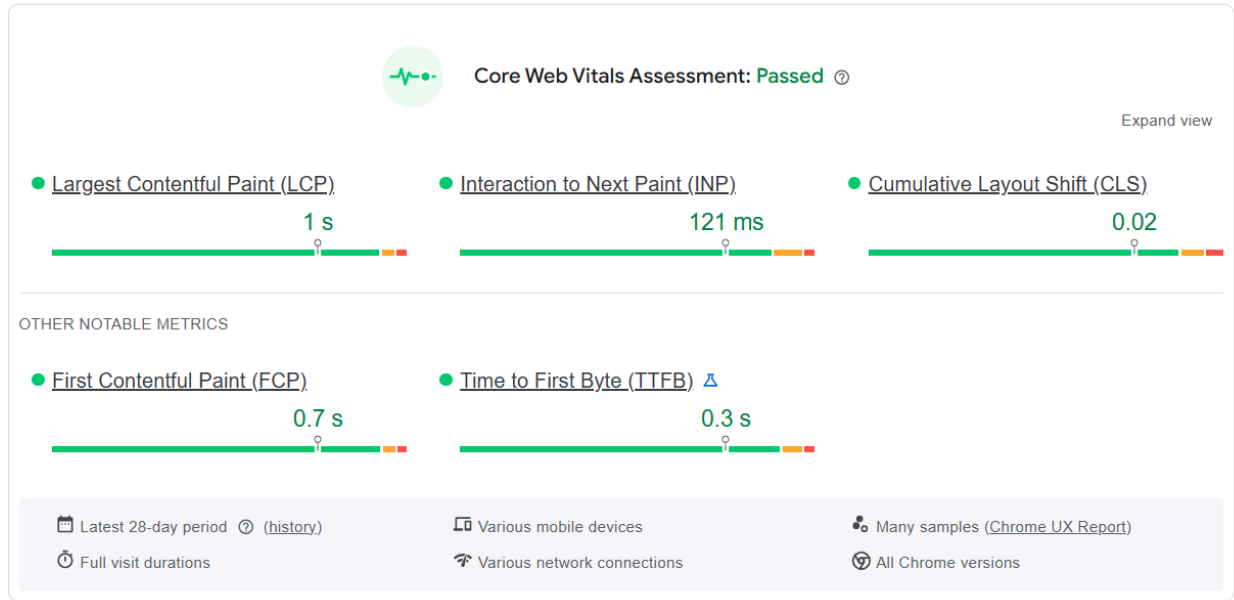
Analyze

Mobile

Desktop

Discover what your real users are experiencing

This URL | Origin



Report from Dec 6, 2025, 12:41:46 AM

https://www.bbc.com/

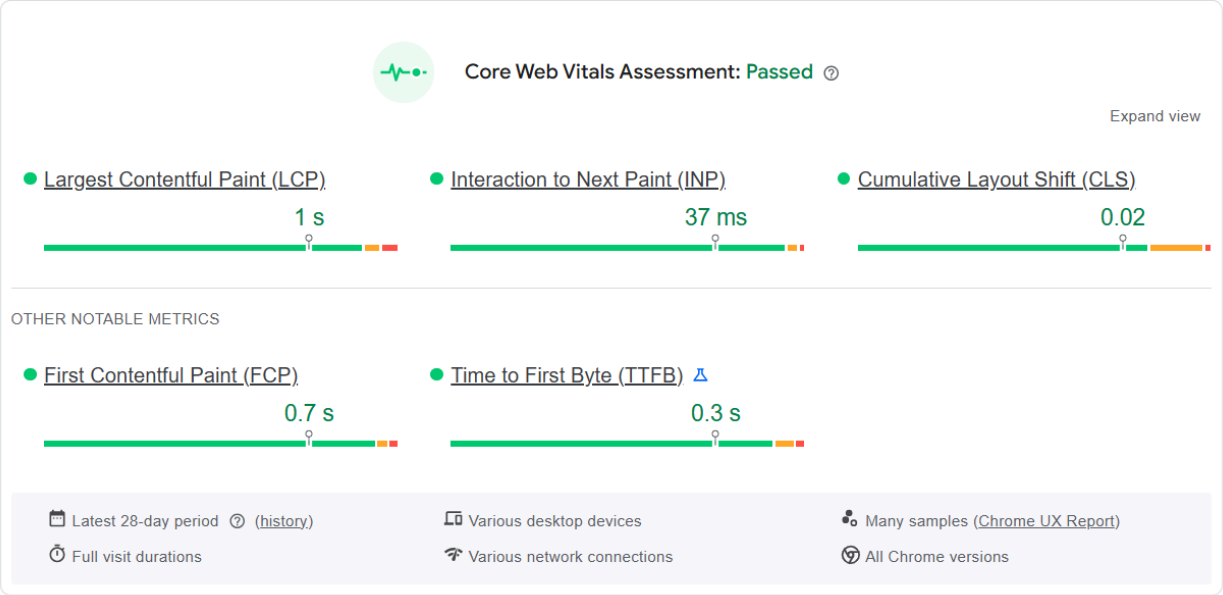
Analyze

Mobile

Desktop

Discover what your real users are experiencing

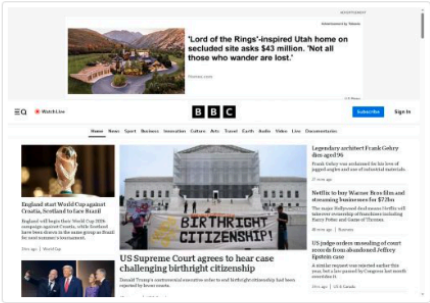
This URL Origin





Performance

Values are estimated and may vary. The [performance score is calculated](#) directly from these metrics. [See calculator.](#)



METRICS

Expand view

● First Contentful Paint
0.9 s

▲ Total Blocking Time
2,760 ms

■ Largest Contentful Paint
1.9 s

● Cumulative Layout Shift
0.01

INSIGHTS

 Use efficient cache lifetimes — Est savings of 1,657 KiB	▼
 Legacy JavaScript — Est savings of 84 KiB	▼
 Font display — Est savings of 40 ms	▼
 Forced reflow	▼
 LCP request discovery	▼
 Network dependency tree	▼
 Improve image delivery — Est savings of 674 KiB	▼
 Layout shift culprits	▼
 Optimize DOM size	▼
 LCP breakdown	▼
 3rd parties	▼

These insights are also available in the Chrome DevTools Performance Panel - [record a trace](#) to view more detailed information.

DIAGNOSTICS

 Reduce JavaScript execution time — 7.1 s	▼
 Minimize main-thread work — 9.7 s	▼

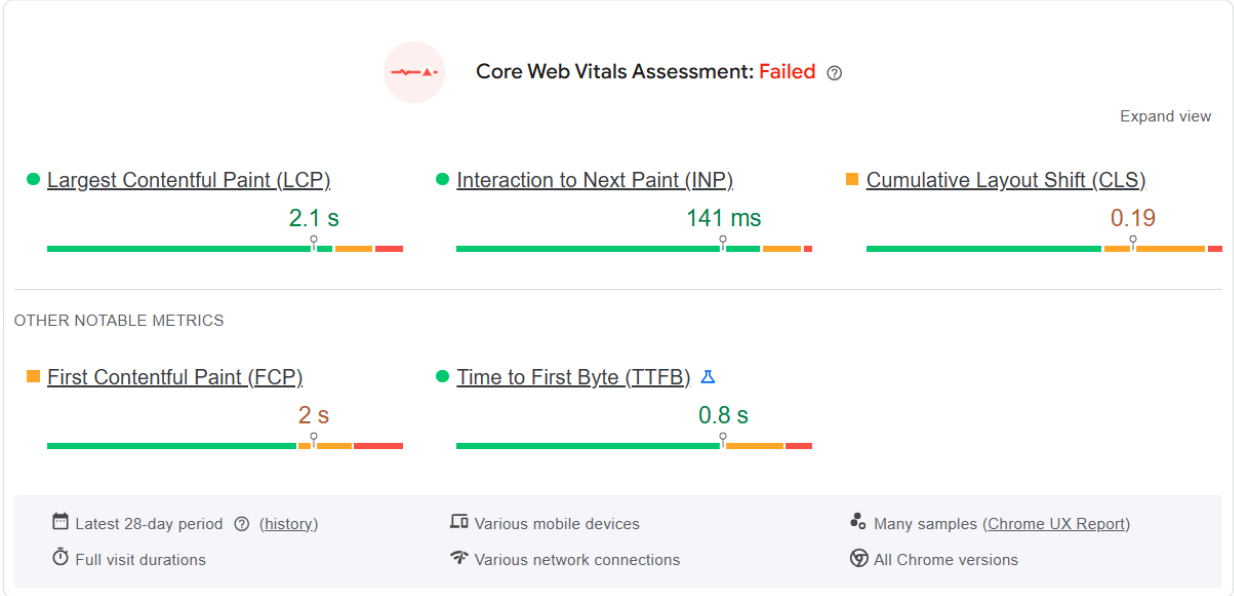
https://web.mit.edu/

Analyze

Mobile Desktop

Discover what your real users are experiencing

This URL Origin



Report from Dec 6, 2025, 12:41:49 AM

https://web.mit.edu/

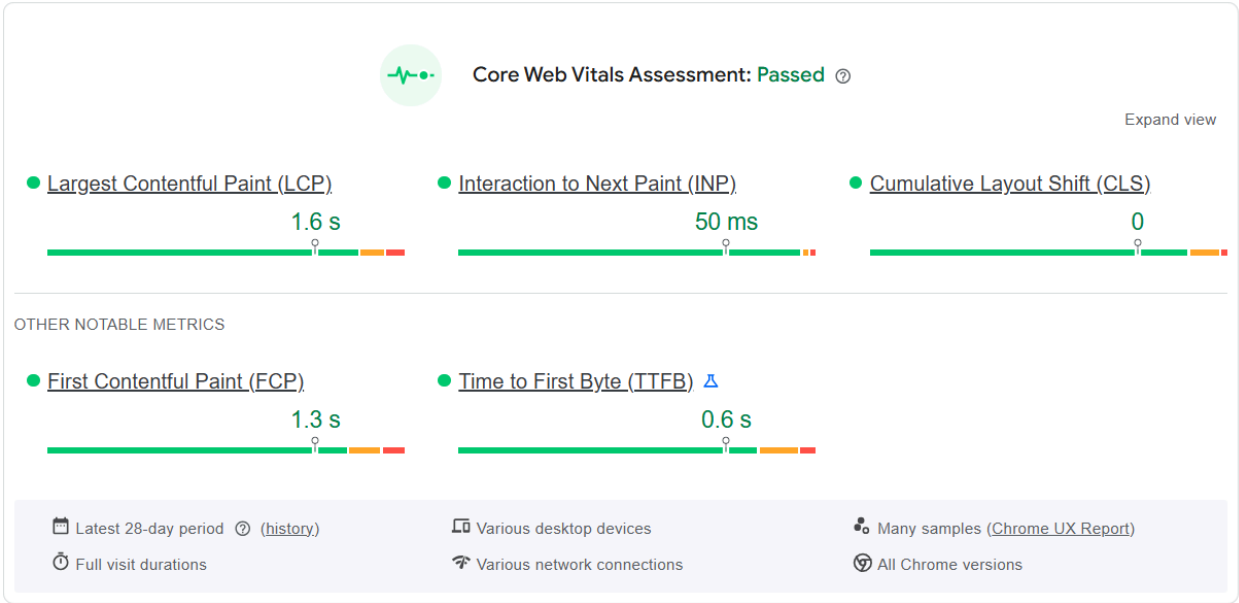
Analyze

Mobile

Desktop

Discover what your real users are experiencing

This URLOrigin





Performance



Accessibility



Best Practices



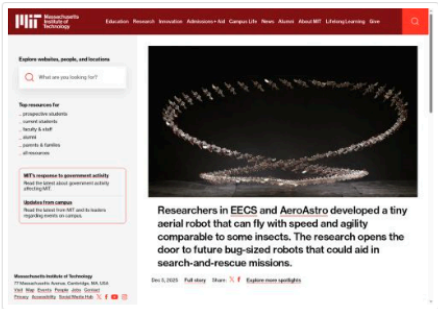
SEO



Performance

Values are estimated and may vary. The [performance score](#) is calculated directly from these metrics. [See calculator.](#)

▲ 0–49 ■ 50–89 ● 90–100



METRICS

Expand view

● First Contentful Paint
0.8 s

● Total Blocking Time
0 ms

■ Largest Contentful Paint
1.3 s

● Cumulative Layout Shift
0.001

INSIGHTS

▲ Render blocking requests — Est savings of 530 ms	▼
▲ Use efficient cache lifetimes — Est savings of 983 KiB	▼
▲ Improve image delivery — Est savings of 353 KiB	▼
▲ Network dependency tree	▼
■ Font display — Est savings of 20 ms	▼
○ Layout shift culprits	▼
○ LCP breakdown	▼
○ 3rd parties	▼

These insights are also available in the Chrome DevTools Performance Panel - [record a trace](#) to view more detailed information.

DIAGNOSTICS

▲ Reduce unused JavaScript — Est savings of 239 KiB	▼
▲ Reduce unused CSS — Est savings of 66 KiB	▼
■ Image elements do not have explicit <code>width</code> and <code>height</code>	▼

More information about the performance of your application. These numbers don't [directly affect](#) the Performance score.